

**To:**

Estelle Altazin, Senior Data Scientist

**Subject:**

Approach to Investigating Customer Churn at PowerCo

Hi Estelle,

Following our discussion on the PowerCo case, I have outlined the data requirements and the analytical approach I propose for investigating the drivers of customer churn.

To assess whether price sensitivity is a key factor influencing churn, we should request customer-level data including historical electricity and gas prices, billing information, contract type and duration, energy consumption, customer tenure, churn status, customer service interactions, payment history, location, and relevant demographic or business information. If available, competitor pricing data would also provide valuable market context.

Once the data is available, I will follow a structured data science approach. I will begin with exploratory data analysis (EDA) and data cleaning to understand the dataset, address missing values, and identify trends or anomalies. Next, I will perform feature engineering by creating variables such as price sensitivity, usage patterns, and customer tenure. To test the hypothesis, I will build and compare binary classification models, such as Logistic Regression, Decision Trees, and Random Forest, to predict customer churn. Model performance will be evaluated using metrics including Accuracy, Precision, Recall, F1-score, and ROC-AUC, and feature importance analysis will be used to quantify the impact of price sensitivity relative to other factors.

The insights from this analysis will help determine whether price sensitivity is the primary driver of churn or whether other customer or market factors have a greater influence, enabling us to provide actionable recommendations to improve customer retention.

Kind regards,

**Simerjeet Kaur**

Junior Data Scientist